The tomato leafminer across Asia

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Table of contents

1 Overview 1

2 The climate 2

3 The invasion 3

4 The timing against the records 4

5 How far the moth flies in a month 5

6 Deviations from the source 5

1 Overview

The tomato leafminer *Tuta absoluta* left South America for Spain in 2006, crossed Africa within a decade, and was first found in Asia, in Turkey, in 2009. By 2018 it was in India, Nepal, Bangladesh, Central Asia and probably Myanmar. Could it have spread that fast by flying? Guimapi et al. (2020) asked with the cellular automaton that Guimapi et al. (2016) had built for Africa:

- the land is cut into cells, and time into months;
- a cell next to the invasion becomes invaded in a month whose mean temperature is at least 22 °C and mean relative humidity at least 55%, and exposed otherwise: the moth may cross it but cannot establish;
- an exposed cell becomes invaded in a later suitable month;
- exposed and invaded cells both carry the invasion on.

Their answer was no: flight alone would take about 20–25 years to reach Southeast Asia, so the pest must have travelled with traded tomatoes and people.

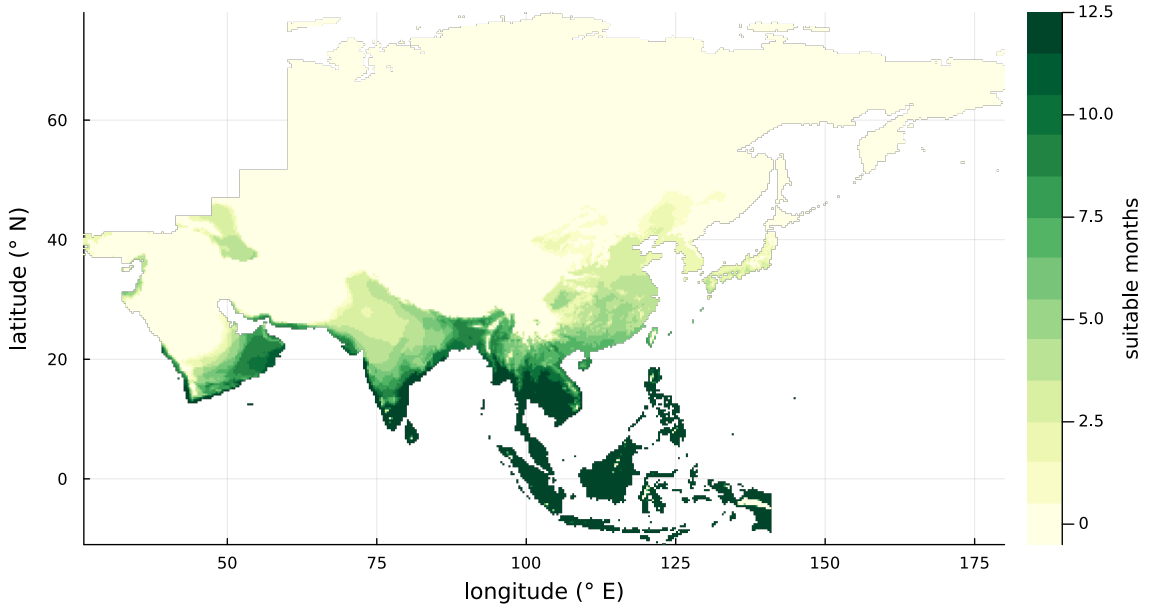
This implementation is an adaptation. The authors' code and data are not published, so the climate is the CRU CL 2.0 climatology (New et al. 2002) on cells of 1/3 degree, bundled with the package, and the reach of a month is set from the timing of the source's maps rather than from its text, as explained below.

```
ENV["GKSwstype"] = "100"
using EcologicalCellularAutomata
using CellularAutomataCore
using Plots
```

2 The climate

The bundled climate holds the monthly mean temperature and relative humidity of each land cell of Asia. A cell can be invaded only in a month that meets both thresholds. Much of Asia never does: the north is too cold, and the deserts and steppes of the west and the interior are too dry in the months that are warm enough.

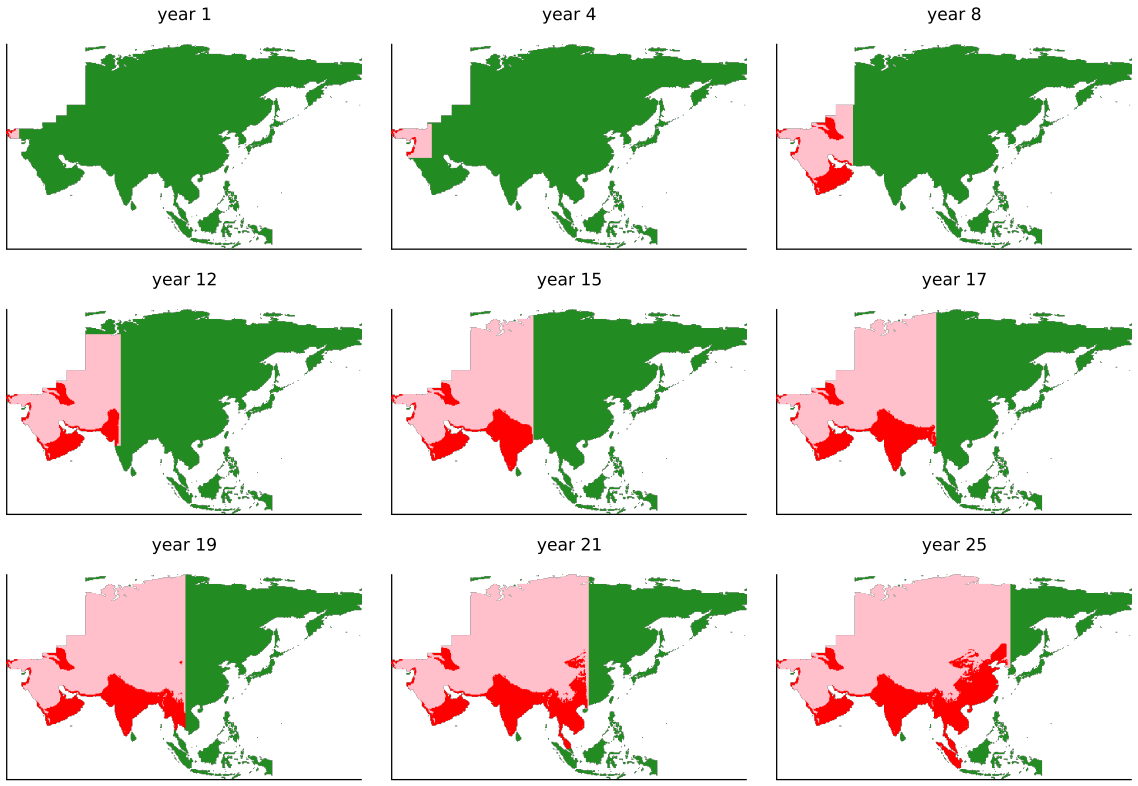
```
c = tuta_climate()
months = dropdims(sum((c.temperature .>= 22) .& (c.humidity .>= 55); dims=3);
dims=3)
suitable = [c.land[I] ? Float64(months[I]) : NaN for I in CartesianIndices(c.land)]
lats = reverse(c.lat)                    # rows run from the north; plots want
rising latitude
northup(M) = reverse(M; dims=1)
heatmap(c.lon, lats, northup(suitable); c=cgrad(:YlGn, 13; categorical=true),
clims=(-0.5, 12.5),
        aspect_ratio=1, xlims=(26, 180), ylims=(-11, 78), xlabel="longitude (° E)",
        ylabel="latitude (° N)", colorbar_title="suitable months", size=(850, 480),
        left_margin=3Plots.mm, bottom_margin=3Plots.mm)
```



3 The invasion

The run starts in January from İzmir, near the first record in Turkey, and lasts 25 years. Each cell records the month in which it was first reached and the month in which it was invaded, so one run gives the maps of every year.

```
prob = problem(:tuta_absoluta_asia)
u = solve(prob; save_everystep=false).u[end]
function snapshot(u, year)
    m = 12 * year
    return [!u.land[I] ? NaN :
            0 <= u.invaded[I] <= m ? 2.0 : 0 <= u.reached[I] <= m ? 1.0 : 0.0
            for I in CartesianIndices(u.land)]
end
colours = cgrad([:forestgreen, :pink, :red], 3; categorical=true)
plot([heatmap(c.lon, lats, northup(snapshot(u, y)); c=colours, clims=(-0.5, 2.5),
colorbar=false,
              aspect_ratio=1, xlims=(26, 180), ylims=(-11, 78), title="year $y",
              titlefontsize=9, ticks=false)
      for y in (1, 4, 8, 12, 15, 17, 19, 21, 25)]...; layout=(3, 3), size=(900,
620))
```



Green cells are susceptible, pink exposed and red invaded. As in the source's Fig. 2, the invasion crosses Anatolia in its first four years and the Levant, Iraq and Arabia by its eighth, establishing

only in the south-west of Arabia; it crosses Iran and Central Asia as exposed land, reaches the Indian subcontinent after 12–15 years and Indochina, southern China and the Malay peninsula after 17–21. Korea is reached overland only in the last months; Taiwan, the Philippines and Japan are never reached, as the sea is too wide to cross. The source's maps colour every cell the invasion has reached, exposed or invaded, in red. They leave northern Asia, Mongolia and Tibet untouched, where here the invasion crosses them as exposed land; the deviations below explain why.

4 The timing against the records

```
places = [("India", 2015, 28.6, 77.2), ("Nepal", 2016, 27.7, 85.3),
          ("Bangladesh", 2016, 23.8, 90.4), ("Kyrgyzstan", 2017, 42.9, 74.6),
          ("Tajikistan", 2018, 38.6, 68.8), ("Myanmar", 2018, 16.8, 96.2),
          ("Thailand", nothing, 13.8, 100.5), ("Vietnam", nothing, 21.0, 105.8),
          ("southern China", nothing, 23.1, 113.3), ("Malaysia", nothing, 3.1,
101.7),
          ("Taiwan", nothing, 23.7, 121.0), ("Philippines", nothing, 14.6, 121.0)]
yrs(x) = x === nothing ? missing : round(x; digits=1)
[(; place, recorded=year === nothing ? missing : year - 2009,
  reached=yrs(tuta_arrival(u, lat, lon).reached),
  invaded=yrs(tuta_arrival(u, lat, lon).invaded))
 for (place, year, lat, lon) in places]
```

```
12-element Vector{NamedTuple{(:place, :recorded, :reached, :invaded)}}:
 (place = "India", recorded = 6, reached = 12.5, invaded = 12.6)
 (place = "Nepal", recorded = 7, reached = 14.5, invaded = 14.5)
 (place = "Bangladesh", recorded = 7, reached = 15.8, invaded = 15.8)
 (place = "Kyrgyzstan", recorded = 8, reached = 11.8, invaded = missing)
 (place = "Tajikistan", recorded = 9, reached = 10.4, invaded = missing)
 (place = "Myanmar", recorded = 9, reached = 17.5, invaded = 17.5)
 (place = "Thailand", recorded = missing, reached = 18.3, invaded = 18.3)
 (place = "Vietnam", recorded = missing, reached = 19.7, invaded = 19.7)
 (place = "southern China", recorded = missing, reached = 21.5, invaded = 21.5)
 (place = "Malaysia", recorded = missing, reached = 20.9, invaded = 20.9)
 (place = "Taiwan", recorded = missing, reached = missing, invaded = missing)
 (place = "Philippines", recorded = missing, reached = missing, invaded = missing)
```

Years after 2009 at which each place was recorded and at which the simulation first reaches and invades it (at a city: Delhi, Kathmandu, Dhaka, Bishkek, Dushanbe, Yangon, Bangkok, Hanoi, Guangzhou, Kuala Lumpur, central Taiwan, Manila). Flight alone is about twice as slow as the pest was: it reaches India in 12.6 years against 6, and Myanmar in 17.5 against 9, as the source concluded. Kyrgyzstan and Tajikistan, where the pest was found, are reached but never invaded here: they are too dry in their warm months.

5 How far the moth flies in a month

The source's text gives the moth's flight as 100 km a month, through a Moore neighbourhood of radius 100 km. That is the reach of its Africa model (Guimapi et al. 2016), whose 25 km cells and 100 km neighbourhood carried the invasion from Spain to Kenya in seven years. In Asia, a reach of about 100 km (three cells here) is far faster than the source's maps:

```
fast = solve(problem(:tuta_absoluta_asia; radius=3); save_everystep=false).u[end]
[(; place, radius_1=yrs(tuta_arrival(u, lat, lon).reached),
  radius_3=yrs(tuta_arrival(fast, lat, lon).reached))
 for (place, _, lat, lon) in places]
```

```
12-element Vector{NamedTuple{(:place, :radius_1, :radius_3)}}:
 (place = "India", radius_1 = 12.5, radius_3 = 4.2)
 (place = "Nepal", radius_1 = 14.5, radius_3 = 4.8)
 (place = "Bangladesh", radius_1 = 15.8, radius_3 = 5.3)
 (place = "Kyrgyzstan", radius_1 = 11.8, radius_3 = 4.0)
 (place = "Tajikistan", radius_1 = 10.4, radius_3 = 3.5)
 (place = "Myanmar", radius_1 = 17.5, radius_3 = 5.8)
 (place = "Thailand", radius_1 = 18.3, radius_3 = 6.2)
 (place = "Vietnam", radius_1 = 19.7, radius_3 = 6.6)
 (place = "southern China", radius_1 = 21.5, radius_3 = 7.2)
 (place = "Malaysia", radius_1 = 20.9, radius_3 = 7.0)
 (place = "Taiwan", radius_1 = missing, radius_3 = 7.9)
 (place = "Philippines", radius_1 = missing, radius_3 = 10.9)
```

With a reach of about 100 km the invasion is in India after four years and in Indochina after six, and it crosses to Taiwan, the Philippines and Japan, where the source finds no risk of natural spread. The source's maps match a reach of one cell a month: one of its 25 km cells, or one cell of 1/3 degree (about 37 km) here.

6 Deviations from the source

```
foreach(d -> println("- ", d), model_info(:tuta_absoluta_asia).deviations)
```

```
- The authors' code and data are not published. The climate is the CRU CL 2.0
climatology (monthly mean temperature and relative humidity, 1961-1990; New et al.
2002) on cells of 1/3 degree, bundled under the Open Database License, rather than
WorldClim temperature at 1 km and relative humidity from NASA SSE interpolated from
1 degree by inverse distance weighting.
- The rule is that of Guimapi et al. (2016), which the source applies without its
vegetation index and tomato production: a land cell reached by an exposed or invaded
Moore neighbour becomes invaded in a month that meets both thresholds and exposed
otherwise, and exposed cells carry the invasion on and become invaded in a later
suitable month. The source's text states only the change from susceptible to
invaded; read alone, the invasion stalls in western Turkey, where no month is both
warm and humid enough.
```

```
- The reach in a month is one cell of 1/3 degree (about 37 km), which matches the
timing of the source's maps, like one of the 25 km cells of Guimapi et al. (2016);
the source's text gives a Moore neighbourhood of radius 100 km (radius = 3 here),
which spreads the invasion two to three times as fast and across the seas to Taiwan,
the Philippines and Japan.
- The thresholds are inclusive (at least 22 degC and 55%), as in Guimapi et al.
(2016); the source's text says greater than.
- The study area is the land of Asia within a rough boundary (no Europe west of the
Turkish straits, the Black Sea, the Caucasus and the Ural river and mountains; no
Africa west of Suez and the Red Sea, nor the Horn of Africa; no New Guinea); a cell
is land if at least two of its four 10' cells are. The invasion starts in January in
the cell of Izmir; the source marks a point in western Turkey and gives no month.
- The source's maps colour every cell the invasion has reached, exposed or invaded,
in red; here the two are told apart. Exposure is not limited by the climate, as in
the temperature-and-humidity runs of Guimapi et al. (2016), where all of Africa is
reached; so here the invasion also crosses northern Asia, Mongolia and Tibet as
exposed land, which the source's maps leave untouched. Limiting exposure to months
of at least 22 degC matches those maps in pattern on cells of 10' with a reach of
about 100 km, but spreads about one and a half times as fast as they do and stalls
on the Anatolian plateau on the cells of 1/3 degree bundled here.
```


Guimapi, Ritter A., Ramasamy Srinivasan, Henri E. Tonnang, Paola Sotelo-Cardona, and Samira A. Mohamed. 2020. "Exploring the Mechanisms of the Spatiotemporal Invasion of Tuta absoluta in Asia." *Agriculture* 10 (4): 124. https://doi.org/10.3390/agriculture10040124.

Guimapi, Ritter Y. A., Samira A. Mohamed, George O. Okeyo, Frank T. Ndjomatchoua, Sunday Ekesi, and Henri E. Z. Tonnang. 2016. "Modeling the Risk of Invasion and Spread of Tuta absoluta in Africa." *Ecological Complexity* 28: 77–93. https://doi.org/10.1016/j.ecocom.2016.08.001.

New, Mark, David Lister, Mike Hulme, and Ian Makin. 2002. "A High-Resolution Data Set of Surface Climate over Global Land Areas." *Climate Research* 21: 1–25. https://doi.org/10.3354/cr021001.